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PES2UG23CS285

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The questions:

1. Based on the metrics and the visualizations, what inferences about the performance of the Linear Kernel can you draw?

Ans: The SVM with a Linear Kernel performs poorly on the Moons dataset because it can only create a straight-line decision boundary. As the visualization clearly shows, this straight line is fundamentally unable to separate the data's curved, crescent shapes, resulting in significant misclassifications. This proves that a linear model is inappropriate for this non-linear problem, leading to lower accuracy scores.

2. Compare the decision boundaries of the RBF and Polynomial kernels. Which one seems to capture the shape of the data more naturally?

Ans: Both the RBF and Polynomial kernels create effective non-linear boundaries. However, the RBF kernel typically captures the shape of the data more naturally. Its boundary is more flexible and localized, creating a smooth curve that precisely follows the crescent shape of the "moons."

3. In this case, which kernel appears to be the most effective?

Ans: The RBF kernel is superior here because, it handles Non-Linearity: While the data (Variance vs. Skewness) appears somewhat separable, the decision boundary is often slightly curved, which the non-linear RBF kernel can model perfectly.

4. The Polynomial kernel shows lower performance here compared to the Moons dataset. What might be the reason for this?

Ans: The polynomial kernel shows lower performance on the Banknote dataset compared to the Moons dataset primarily because of the nature of the decision boundary required.

5. Compare the two plots. Which model, the "Soft Margin" ($C=0.1$) or the "Hard Margin" ($C=100$), produces a wider margin?

Ans: The Soft Margin model ($C=0.1$) produces the wider margin. A smaller C value means the optimization allows for more misclassified points and violations of the margin constraints, resulting in a more robust and broader separation between the classes.

6. Look closely at the "Soft Margin" ($C=0.1$) plot. You'll notice some points are either inside the margin or on the wrong side of the decision boundary. Why does the SVM allow these "mistakes"? What is the primary goal of this model?

Ans: The SVM allows these "mistakes" (points inside or across the margin) because the primary goal of the Soft Margin model is to maximize the margin width while keeping the total classification error below a certain tolerance. The parameter C acts as a penalty for these violations, allowing the model to prioritize generalization over perfect training set accuracy.

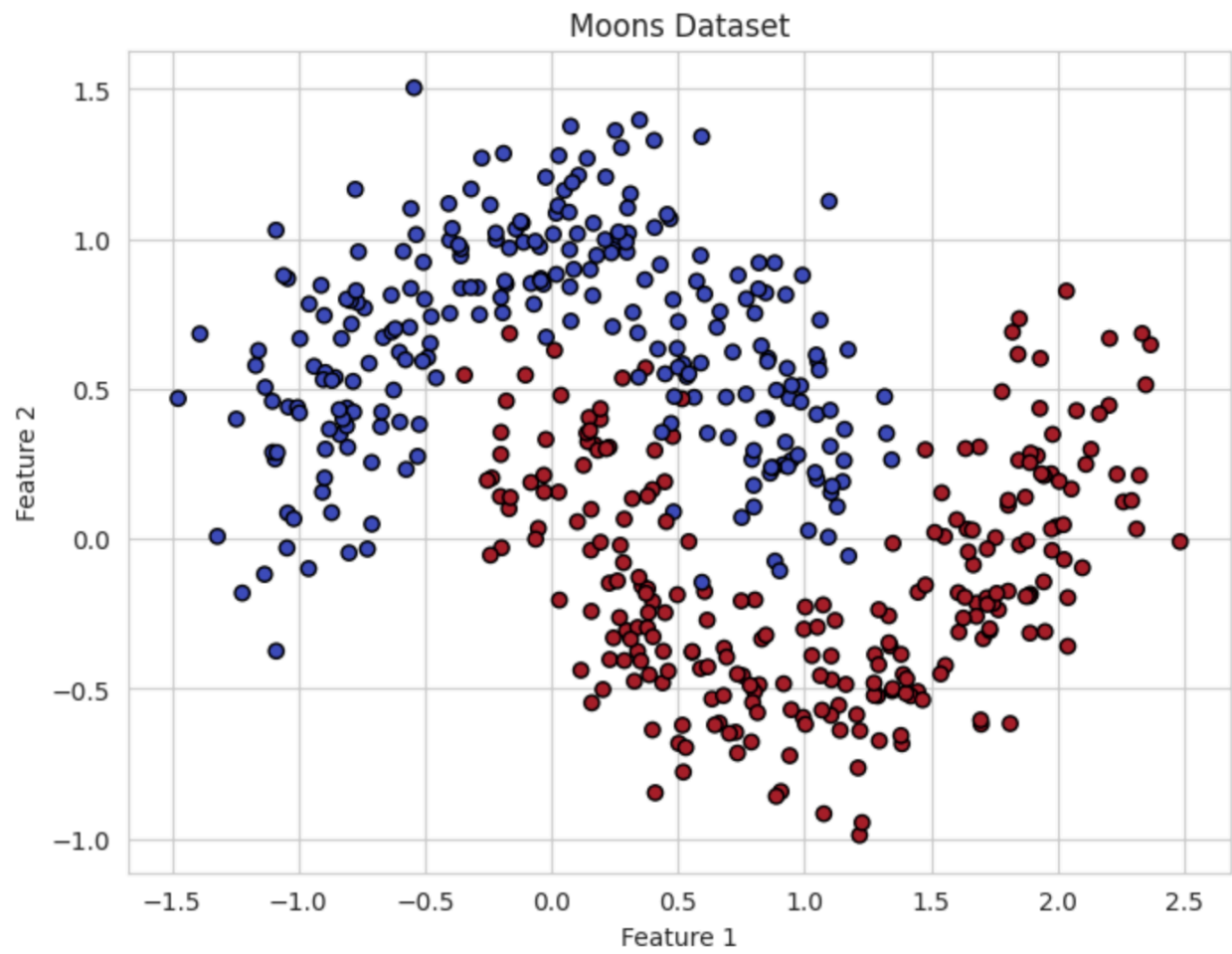
7. Which of these two models do you think is more likely to be overfitting to the training data? Explain your reasoning.

Ans: The Hard Margin model ($C=100$) is more likely to be overfitting. Its large C value harshly penalizes every misclassification, forcing the decision boundary to be narrow and highly sensitive to individual noisy data points or outliers, which won't generalize well to new, unseen data.

8. Imagine you receive a new, unseen data point. Which model do you trust more to classify it correctly? Why? In a real-world scenario where data is often noisy, which value of C (low or high) would you generally prefer to start with?

Ans: I would trust the Soft Margin ($C=0.1$) model more because its wider margin indicates better generalization and robustness to noise. In a real-world scenario where data is often noisy and has outliers, you would generally prefer to start with a low value of C to build a boundary that captures the overall trend, rather than the specific noise.

Visualization of the Moons dataset.



Classification report:

⇒ SVM with LINEAR Kernel <PES2UG23CS285>

	precision	recall	f1-score	support
0	0.85	0.89	0.87	75
1	0.89	0.84	0.86	75
accuracy			0.87	150
macro avg	0.87	0.87	0.87	150
weighted avg	0.87	0.87	0.87	150

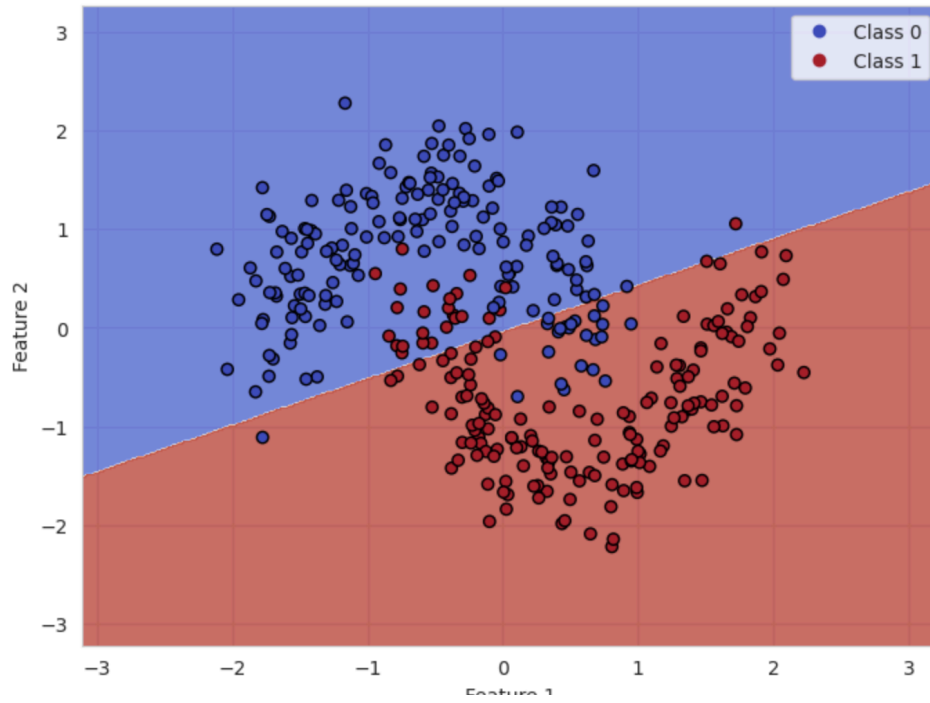
SVM with RBF Kernel <PES2UG23CS285>

	precision	recall	f1-score	support
0	0.95	1.00	0.97	75
1	1.00	0.95	0.97	75
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

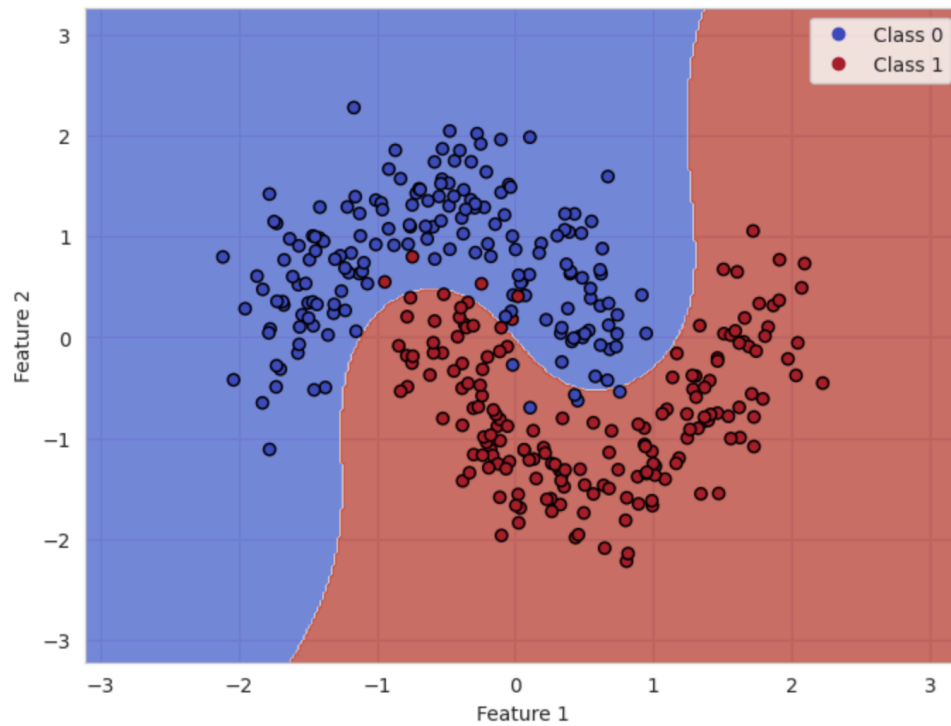
SVM with POLY Kernel <PES2UG23CS285>

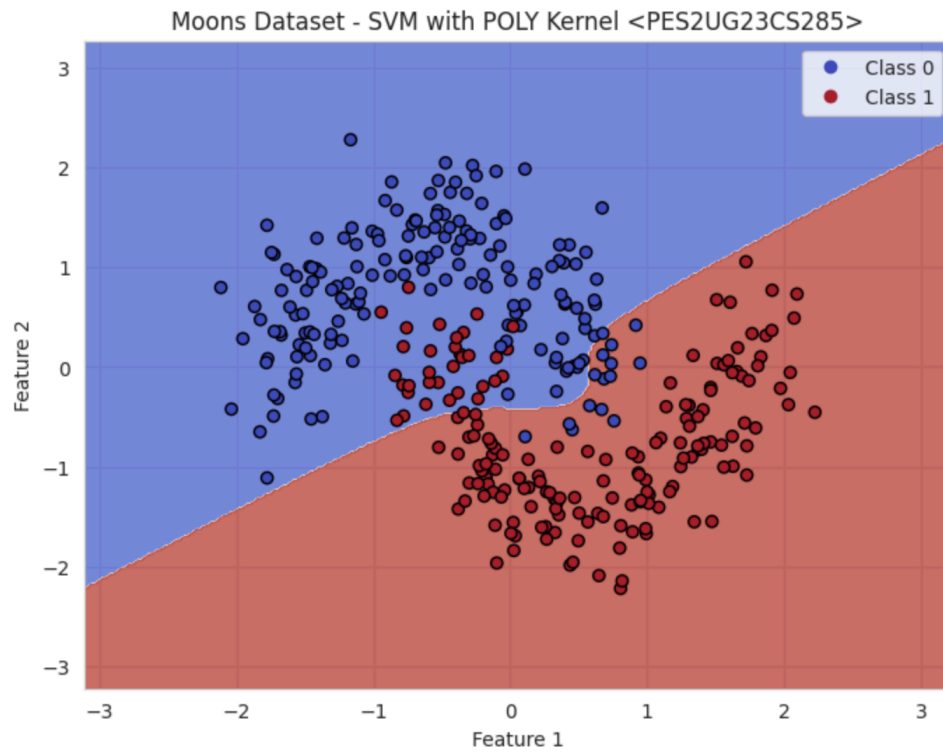
	precision	recall	f1-score	support
0	0.85	0.95	0.89	75
1	0.94	0.83	0.88	75
accuracy			0.89	150
macro avg	0.89	0.89	0.89	150
weighted avg	0.89	0.89	0.89	150

Moons Dataset - SVM with LINEAR Kernel <PES2UG23CS285>

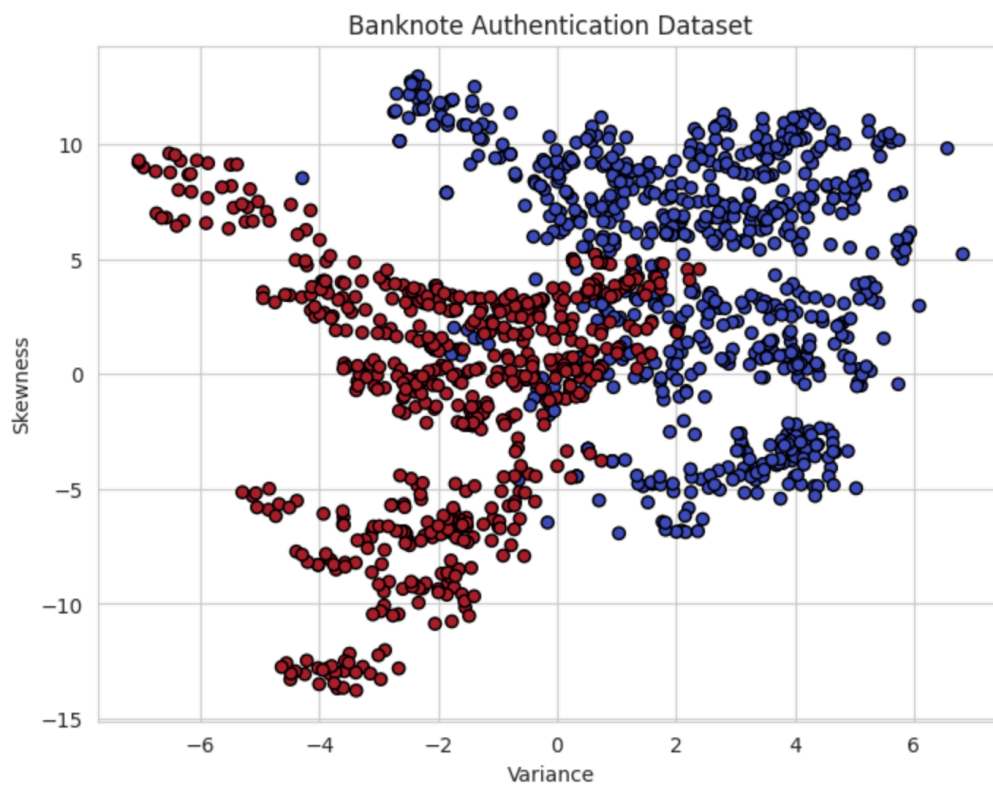


Moons Dataset - SVM with RBF Kernel <PES2UG23CS285>





Banknote dataset visualization.





SVM with LINEAR Kernel <PES2UG23CS285>

	precision	recall	f1-score	support
Forged	0.90	0.88	0.89	229
Genuine	0.86	0.88	0.87	183
accuracy			0.88	412
macro avg	0.88	0.88	0.88	412
weighted avg	0.88	0.88	0.88	412

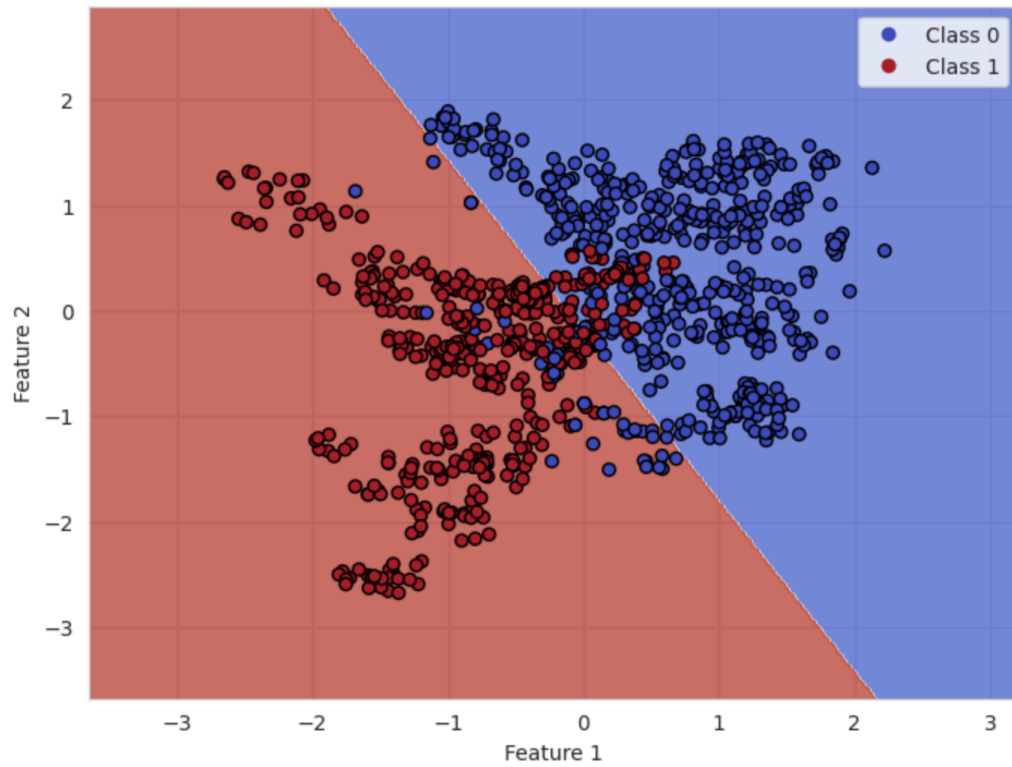
SVM with RBF Kernel <PES2UG23CS285>

	precision	recall	f1-score	support
Forged	0.96	0.91	0.94	229
Genuine	0.90	0.96	0.93	183
accuracy			0.93	412
macro avg	0.93	0.93	0.93	412
weighted avg	0.93	0.93	0.93	412

SVM with POLY Kernel <PES2UG23CS285>

	precision	recall	f1-score	support
Forged	0.82	0.91	0.87	229
Genuine	0.87	0.75	0.81	183
accuracy			0.84	412
macro avg	0.85	0.83	0.84	412
weighted avg	0.85	0.84	0.84	412

Banknote Dataset - SVM with LINEAR Kernel <PES2UG23CS285>



Banknote Dataset - SVM with RBF Kernel <PES2UG23CS285>

